

TITLE PAGE

- Problem Statement ID – SIH26001
- Problem Statement Title - AI-Based early warning and landslide Risk Monitoring System in NER
- Theme - Disaster Management
- PS Category - Software
- Team ID- SIH26-A0H-T327
- Team Name (Registered on portal) - Solution Squad



Proposed Solution

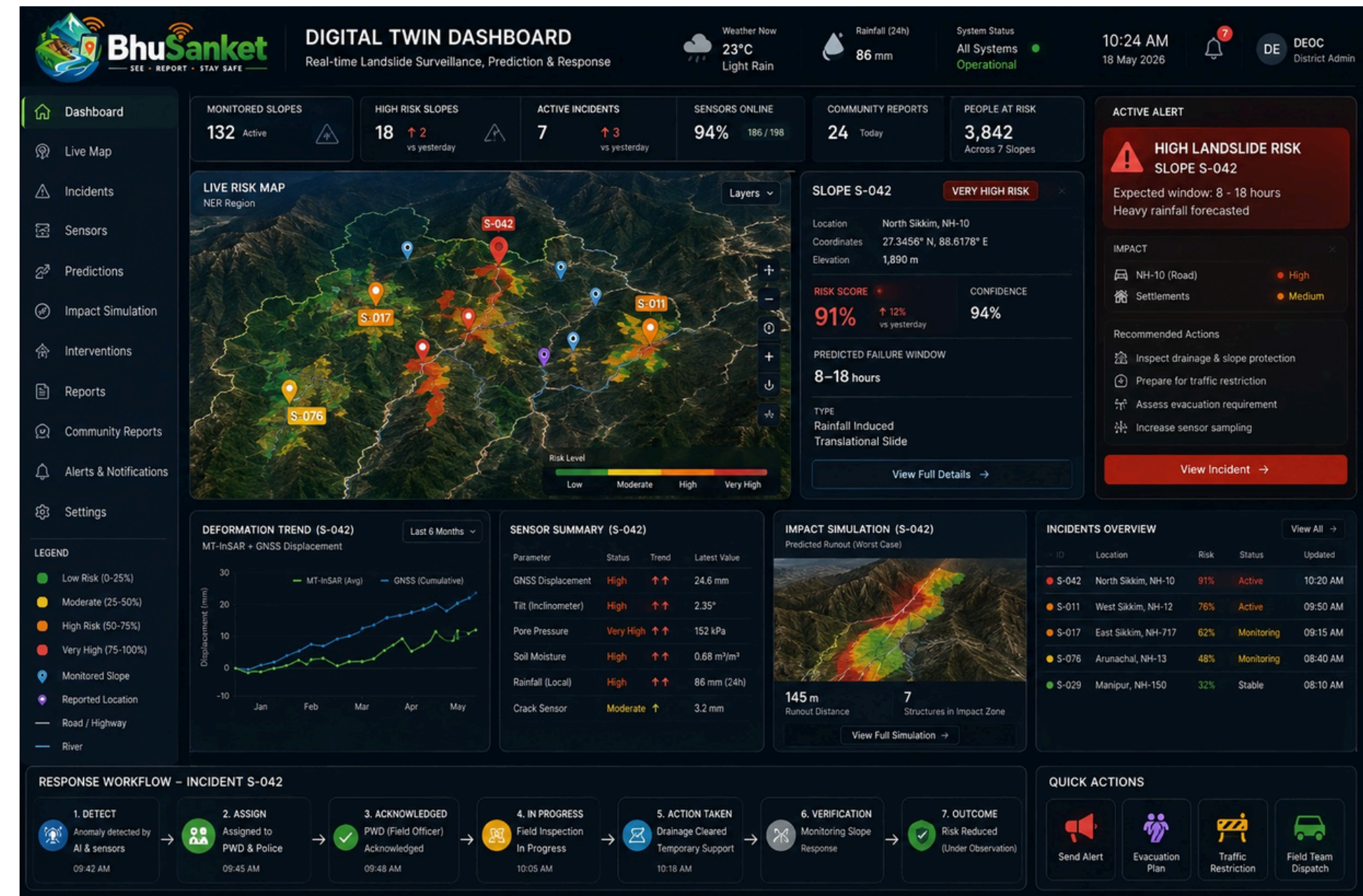
- AI-powered landslide intelligence-to-action platform for NER
- Fuses MT-InSAR + IoT sensors + weather + community data
- Builds a live Digital Twin of vulnerable slopes
- Predicts landslide risk, deformation and time-to-failure
- Simulates impact and potential interventions
- Converts risk into prioritized incidents and authority actions
- Enables field response, escalation and intervention verification
- Creates a closed loop: Predict → Act → Verify

PROBLEM ADDRESSED

- Delayed response from warning to field action
- Fragmented monitoring across data sources
- Limited ground visibility in remote NER terrain
- Reactive mitigation after landslide occurrence
- Coordination & accountability gaps
- Limited post-intervention feedback

What Makes It Unique

- Multi-Source — Satellite + IoT + Community
- Adaptive — Sensors respond to anomalies
- AI-to-Action — Risk → Assigned action → Escalation
- Closed-Loop — Predict → Act → Verify → Learn



MT-InSAR → IoT → Fusion → AI → Simulation → Authority → | | Field Action → Verification

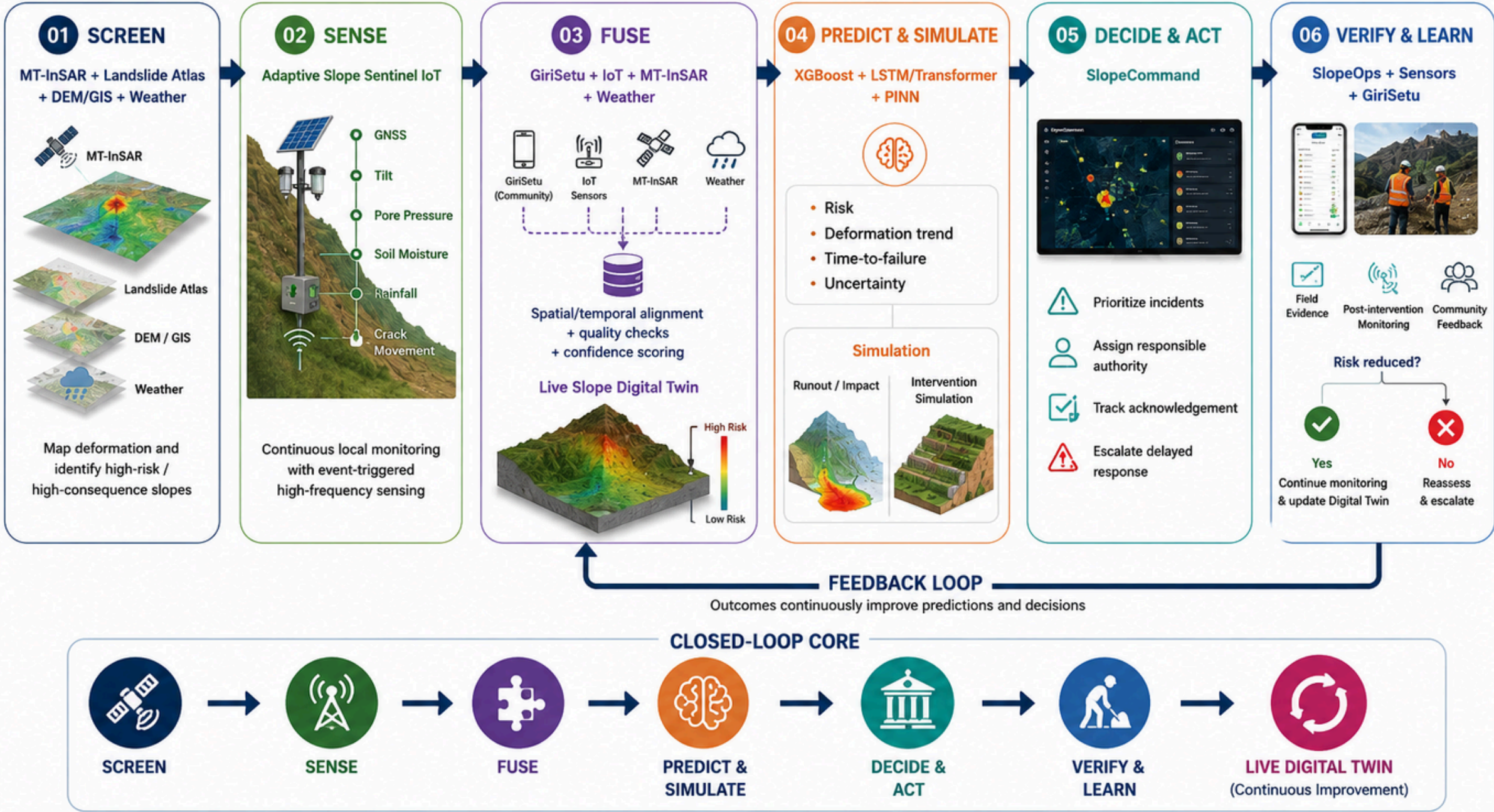
Technologies:

Layer	Technologies
AI / ML	Python, PyTorch, TensorFlow, Scikit-
Geo-Spatial	MT-InSAR, Google Earth Engine, QGIS,
IoT / Communication	ESP32, LoRaWAN, MQTT
Backend / Database	FastAPI, Node.js, PostgreSQL
Mobile	Flutter, Firebase, Google Maps
Cloud	AWS
Data Sources	ISRO, NRSC, GSI, IMD



Report Hazards, Receive Alerts, Stay Safe
 Citizen Reporting, Alerts, Location, Offline Safety

Methodology and process for implementation :



Feasibility:

- Uses existing satellite, IoT, GIS and AI technologies.
- MT-InSAR enables wide-area screening; sensors are deployed only at priority slopes.
- Prototype feasible with ESP32 sensors + cloud backend + AI models + mobile apps.
- Designed to integrate with existing government disaster-response infrastructure.

Challenges → Solutions

- Remote terrain & connectivity → LoRa/4G + offline-first apps
- Sensor/InSAR uncertainty → Multi-source validation + confidence scoring
- False alarms → AI + physics + human verification
- Delayed response → Task tracking + automated escalation
- High deployment cost → Risk-based, adaptive sensor deployment
- Model drift → Continuous field feedback + Digital Twin updates



Regional Monitoring

MT-InSAR + Weather + DEM/GIS

- Find high-risk slopes
- Monitor priority areas

Local Monitoring

Slope Sentinel IoT

- GNSS • Tilt • Moisture • Pressure • Rainfall
- Continuous monitoring

Offline Sync

BhuSanket

- Report with GPS + photo
- Save offline → Sync when online

Impact:

- Citizens: Earlier alerts, safer travel and faster evacuation.
- Authorities: Faster decisions, clear task ownership and coordinated response.
- Field Teams: Real-time risk data for safer inspections and intervention.
- Infrastructure Agencies: Early identification of roads, bridges and critical assets at risk.

Benefits:

UNIQUE IMPACT:

Early Risk → Priority → Assigned Action → Verified Safety

Adaptive: Sensors intensify only when anomalies emerge.

Actionable: Every critical risk can become an assigned incident.

Accountable: Delayed responses trigger escalation.

Learning: Field outcomes update the Digital Twin.



Safer Communities

Earlier warnings, safer evacuation and improved community preparedness.



Faster Action

Converts alerts into assigned tasks and speeds up response with escalation.



Data-Driven Decisions

AI-backed risk scores and Digital Twin enable informed, explainable decisions.



One Response Network

Brings PWD, Police, DDMA, SDRF and other agencies together on one platform.



Continuous Learning

Field observations and intervention feedback update the Digital Twin and improve future predictions.



Lives & Infrastructure Protected

Reduces landslide impact on lives, NER roads, villages and other critical assets.

From Detection → to Decision → to Action → to Verified Safety

Research / Source	Key Contribution	Link
Chutia et al. (2026)	AI-based rainfall-induced landslide prediction in NER	https://doi.org/10.1007/s11069-025-07965-y
ISRO–NRSC (2023)	Landslide Atlas of India; historical inventory & risk baseline	https://www.isro.gov.in/media_isro/pdf/LandslideAtlas_2023.pdf
NIDM	Landslide monitoring, mitigation & community response	https://nidm.gov.in/PDF/modules/Landslide.pdf
Alonso-Díaz et al. (2026)	MT-InSAR + ML for geohazard monitoring	https://doi.org/10.3390/rs18091356
Wang et al. (2026)	Dynamic susceptibility using SBAS-InSAR + XGBoost	https://doi.org/10.3390/rs18040578
Cui et al. (2026)	DEM + GAN for landslide deposition/runout prediction	https://doi.org/10.1016/j.enggeo.2026.108752

NER Prediction + MT-InSAR + Ground Monitoring + Impact Simulation + Mitigation + Community & Authority Response